Merging Consecutive Layers of a Multilingual Language Model: Annealed Absorption and Gradual Layer Flattening on Qwen3

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Abstract

- Task: remove 2–8 of the 28 layers of Qwen3-0.6B-Base (and 4 of Qwen3-1.7B-Base) by merging pairs of consecutive layers, measured in six languages (en, hi, fr, es, de, zh) plus two unseen ones (ja, ru).
- Methods: AAM (keep the better layer of each pair, fade the other out, distill) and PRISM-Shrink (build the exact parallel union of a pair, shrink it gradually to one layer while distilling); compared with drop, a matched drop + distillation control, and LaCo.
- Main results: AAM is the most accurate (+2.3 points over the control at 4 layers on 0.6B, $p = 0.002$; +3.2 on 1.7B; significant on all 3 seeds). PRISM-Shrink has the lowest perplexity damage from 4 layers on (+16.6 % vs +24.2 % for the control, mean of 3 seeds) but no accuracy gain.
- Honest limit: at 4+ layers removed from the 0.6B model all methods are close to random guessing (the best keeps 21 % of the original's above-chance skill); 8-bit quantization saves far more memory with almost no loss.

1 Introduction

- A decoder-only transformer applies L layers in sequence; each adds an update to the token state (residual stream): $h_{i+1} = h_i + f_i(h_i)$. Qwen3-0.6B-Base has $L = 28$ layers (Qwen Team, 2025).
- Removing layers reduces sequential compute; the simplest way is to delete layers whose input and output are similar (angular distance; Gromov et al., 2024; Men et al., 2024).
- Deleting loses what the layer wrote; merging tries to keep the work of both layers of a pair in one layer.
- Evaluation is multilingual because a shared layer must serve every language; damage is measured by perplexity, few-shot accuracy on five benchmarks, and an in-context copying probe.
- Contributions: (1) a controlled multilingual comparison (matched control, paired bootstrap tests, three seeds, competence check); (2) AAM and the finding that the choice of the kept layer drives accuracy; (3) PRISM-Shrink, a gradual, distilled variant of layer flattening (Xu et al., 2026), with ICL-packed training text.

2 Background: from the basics to this work

Language modelling:

- Text is split into tokens (Qwen3 vocabulary: 151,936); each token becomes a vector of size d (1,024 for 0.6B) through an embedding table.
- The model predicts $p(x_t \mid x_{<t})$ and is trained with the cross-entropy $-\frac{1}{T}\sum_t \log p(x_t \mid x_{<t})$; its exponential is the perplexity (lower is better).

One transformer layer (Figure 1):

$$N(x) = \gamma \odot x \,/\, \sqrt{\tfrac{1}{d}\textstyle\sum_j x_j^2 + \epsilon}, \tag{1}$$

$$h' = h + \mathrm{Attn}\big(N(h)\big), \tag{2}$$

$$h_{\mathrm{next}} = h' + \mathrm{MLP}\big(N(h')\big), \tag{3}$$

$$\mathrm{Attn}(x) = \mathrm{softmax}\big(QK^\top/\sqrt{d_h} + M\big)\,V, \tag{4}$$

$$\mathrm{MLP}(x) = W_d\big(\mathrm{SiLU}(W_g x) \odot W_u x\big), \tag{5}$$

where N is RMSNorm, Q, K, V are linear maps of x, M is the causal mask, and W_g, W_u, W_d are the gate, up and down projections; grouped-query attention shares each key-value head between 2 of the 16 query heads.

- The residual stream: each layer only adds an update to h. In the middle of the network the updates are small and neighbouring layers read nearly the same input (cosine 0.96–0.97), which is what makes merging neighbours possible.
- Few-shot prompting: solved examples in the prompt; a base model answers by continuing the

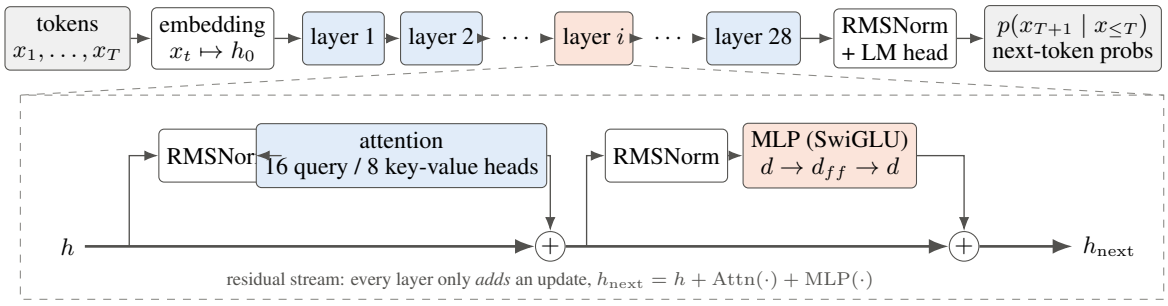


Figure 1: Top: a decoder-only language model turns tokens into vectors, refines them through 28 layers and predicts the next token. Bottom: inside one layer, attention and the MLP each read a normalised copy of the state and *add* their output to the residual stream.

pattern. It relies on *induction heads*, attention heads that find the earlier copy of the current token and predict what followed it (Olsson et al., 2022).

- Knowledge distillation (Hinton et al., 2015): a small or damaged student is trained to match the teacher's output distribution (KL divergence) instead of only the correct token.
- Layer pruning: delete the layers whose input and output are most similar (Gromov et al., 2024; Men et al., 2024); the deleted layer's contribution is lost unless something else learns it.
- Statistics: a bootstrap resamples the test questions to estimate a confidence interval; a paired test compares two models on the same questions (Efron and Tibshirani, 1993).

3 Report

3.1 Feedback and novelties discussed with the mentor

Work completed since Presentation 2 (from the repository history):

- Small test of 15 merging algorithms (closed-form least squares, gradual distillation, search, PRISM family, LaCo), 100 questions per benchmark and language.
- Large test: 200–250 questions per benchmark and language, more than 100k unique test tokens per language, per-question results for bootstrap confidence intervals, and Japanese and Russian as unseen languages.
- A matched control (drop + the same distillation) to separate the effect of a method from the effect of training.
- Depth study (2, 4, 6, 8 layers removed), a second model size (Qwen3-1.7B), a data ablation (kept layer $\times$ training text), a training-length ablation (400 vs 1,000 steps) and three seeds.
- Novelties developed: PRISM (exact parallel form of a layer pair), PRISM-Shrink (gradual shrinking), AAM, ICL-packed training text, CoPlan pair selection, a language-specific residual (LSR), smeared-key merging (SKM) and a cross-term correction (XTC). SKM, XTC and LSR gave no gain (negative results).

3.2 Baselines and state of the art

- Closest state of the art to PRISM-Shrink: FlattenGPT (Xu et al., 2026) merges two adjacent blocks into one wider block and prunes it back to the original width in one step, without training; it reports 87–96 % retained zero-shot performance at about 20 % compression on 7B–70B English models.
- What differs here: gradual, distilled hand-over starting from the exact original pair; ICL-packed training text; multilingual evaluation with a matched control and seeds.
- The control is the most important baseline: it has the same training as the methods, so any gain over it is due to the method itself.

3.3 Datasets

Bias control:

- Language balance: equal calibration text per language, and one sequence per language in every training step.
- No test leakage: calibration and training use Wiki-40B *train*; the pairs to merge are fixed by angular distance on calibration text before any test.
- Held-out calibration sequence per language to measure every fit.
- Competence check: a test cell (benchmark $\times$ language) counts only if the original model is clearly above chance there; only German XNLI on 0.6B is flagged.

Term	Meaning in this report
Token, embedding	a piece of text; its vector of size d
Layer	one transformer block (attention + MLP); Qwen3-0.6B and 1.7B have 28
Residual stream	the running state h of a token; each layer adds its update to it
Attention head, KV group	one attention pattern; a key-value head with the 2 query heads that share it
MLP neuron	one of the d_{ff} hidden units of the feed-forward block
Perplexity	exp of the average next-token loss; how unsure the model is (lower is better)
Few-shot accuracy	share of correct answers when the prompt contains solved examples
Random guessing (chance)	25 % on 4-option tests, 33.3 % on XNLI, 0 % on MGSM; 21.7 % on the average
Skill kept	share of the original's above-chance accuracy that survives (0 = guessing, 100 = original)
Copying loss	loss on the second copy of a repeated random sequence; tests induction heads
Distillation (KD)	training a student to match a teacher's next-token distribution
Angular distance	angle between the state entering and leaving a pair; small = the pair changes little
Drop	delete one layer of each pair, no training
Control	drop + the same distillation as the methods (isolates the method's own effect)
AAM	keep the better layer of each pair, fade the other out, distill
PRISM-Shrink	run the pair as one wide parallel layer, shrink it gradually to one layer, distill
Bootstrap CI, paired p	range that contains the result 95 % of the time; chance of a difference this large if none existed

Table 1: Glossary, from basic to advanced.

Method	Idea	Training	Code	Used here
Drop / ShortGPT (Men et al., 2024; Gromov et al., 2024)	delete the most similar layers	none	simple	yes (`drop`)
LaCo (Yang et al., 2024)	add other layers' weight differences to one layer	none	re-implemented	yes (small test)
Layer parallelism (Calvo González et al., 2025)	run two layers side by side	optional	public (GitHub)	idea only
LLM-Streamline (Chen et al., 2024)	train a small module to replace removed layers	yes	not checked	no
FuseGPT (Pei et al., 2024)	fold removed blocks into neighbours	yes	not checked	no
FlattenGPT (Xu et al., 2026)	one wide layer from two, pruned back in one step	optional	not released	closest prior work
Control (ours)	drop + the same distillation as the methods	yes	this repo	yes

Table 2: Baselines and related depth-compression methods.

Use	Data	Size per language
Calibration	Wiki-40B train (Guo et al., 2020)	16 × 256 tokens
Distillation	Wiki-40B train	300 × 256 tokens
Perplexity test	mC4 (Xue et al., 2021), Wiki-40B test	128 × 256 each
Knowledge	MMMLU (OpenAI, 2024; Hendrycks et al., 2021)	200 (5-shot)
Regional exams	INCLUDE (Romanou et al., 2024)	200 (5-shot)
Maths	MGSM (Shi et al., 2022)	250 (4-shot CoT)
Reading	Belebele (Bandarkar et al., 2023)	200 (3-shot)
Inference	XNLI (Conneau et al., 2018)	200 (3-shot)

Table 3: Data. Tests are never used for fitting or choosing anything. MGSM has no Hindi; INCLUDE has no English.

- Unseen languages (ja, ru) are only evaluated, never used, to detect damage outside the training languages.
- Same shots and prompts for every model; shots never overlap test items.

Bottlenecks: GPU time (Kaggle T4, weekly quota) and GPU memory (the double-width layer in fp32 ran out of memory at 6+ layers and on 1.7B); Hindi needs 2–4× more tokens per item; MMMLU has no development split (the first shuffled rows are used as shots).

Extension: small test (100 questions per benchmark and language) → large test (200–250 questions, 128 instead of 16 perplexity sequences, per-question results, ja and ru added; more than 100k unique test tokens per language, Table 13 in the appendix). ICL-packed training text: each sequence is $[a, b, a]$ (a passage, a passage in another language, the first passage again; 768 tokens), which trains in-context copying.

3.4 Experimental plan

3.4.1 Setting, training and robustness

Pairs are chosen by angular distance $d(i) = \frac{1}{\pi} \arccos \cos(h_i, h_{i+2})$ averaged over tokens (Gromov et al., 2024), never the first or last two layers: 2 removed → 12-13, 14-15; 4 removed → 10-11, 12-13, 14-15, 16-17; 6 adds 7-8, 24-25; 8 adds 5-6, 22-23 (Qwen3-0.6B).

Shared distillation (Hinton et al., 2015): the original model is the frozen teacher; only the merged layers train.

$$\mathcal{L} = \mathcal{L}_{\mathrm{KL}} + \mathcal{L}_h, \tag{6}$$

$$\mathcal{L}_{\mathrm{KL}} = \tfrac{1}{T} \textstyle\sum_t \mathrm{KL}\big(p_t^{\mathrm{tea}} \,\|\, p_t^{\mathrm{stu}}\big), \tag{7}$$

$$\mathcal{L}_h = \|h^s - h^t\|^2 / \|h^t\|^2, \tag{8}$$

where p_t is the next-token distribution at position t and h^s, h^t are the student's and teacher's hidden

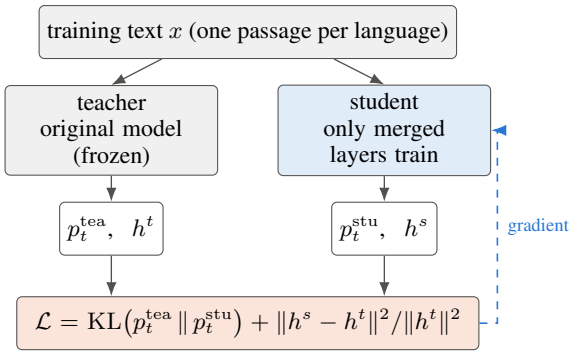


Figure 2: Knowledge distillation shared by the control, AAM and PRISM-Shrink: only the merged layers receive gradients.

states at the output of each merged pair. The removed part is faded out with

$$\alpha(s) = \tfrac{1}{2}\big(1 + \cos(\pi(s+1)/A)\big) \qquad (9)$$

for step s, with $A = 0.7 \times 400 = 280$, then $\alpha = 0$ for the last 120 of 400 steps (Figure 13). At $\alpha = 1$ the student equals the original model exactly (checked before training).

Robustness: identical pairs, data, shots and training recipe for every method; a matched control; paired bootstrap tests on the same questions (Efron and Tibshirani, 1993); three seeds for the main 4-layer runs; a competence check; two model sizes.

3.4.2 Models and approaches

- **Drop**: delete the second layer of each pair. No training.
- **Control**: drop, then the same distillation (the first layer of each pair is kept and trained).
- **AAM**: keep the layer with the lower held-out error when used alone (e.g. pair 14-15: 28.9 % vs 26.6 % $\rightarrow$ keep L15); the other becomes $h + \alpha(f(h) - h)$; distill; remove it.
- **PRISM**: in the middle of the network $h + A(h) + B(h + A(h)) \approx h + A(h) + B(h)$ (input cosine 0.96–0.97). The right side is exactly one wider layer: A's and B's attention head groups and MLP neurons side by side, with each layer's RMSNorm gain folded into its own input weights (exact; checked to $< 10^{-4}$ relative error).
- **PRISM-Shrink**: keep 8 of 16 key-value head groups and 3,072 of 6,144 neurons by write energy; output $= \alpha\, B(A(h)) + (1 - \alpha)\, U_\alpha(h)$, where U_α is the wide layer with every dropped part scaled by α; distill; extract the kept parts as a normal layer.

	Qwen3-0.6B	Qwen3-1.7B
d / d_{ff}	1,024 / 3,072	2,048 / 6,144
n_q / n_{kv} / d_h	16 / 8 / 128	16 / 8 / 128
Layers L	28	28
P_{layer}	15,730,944	50,336,000
Embedding Vd	155,582,464	311,164,928
Total P	596,049,920	1,720,574,976
One layer	2.64 %	2.93 %
4 layers removed	533,126,144	1,519,230,976
Parameters cut	10.6 %	11.7 %

Table 4: Parameter counts (checked against the saved results). 2 / 6 / 8 layers removed from 0.6B: 5.3 / 15.8 / 21.1 %.

Setting	Value
Optimizer	AdamW, lr 5×10^{-5}, no weight decay
Gradient clipping	1.0
Steps	400 (1,000 in the ablation)
Fade schedule	cosine over 280 steps, then 120 at 0
Batch per step	6×256 tokens (ICL: 6×768)
Loss	KL + 1.0 $\times$ hidden MSE
Precision	fp16 autocast (1.7B teacher in fp16)
Micro-batch	1,536 tokens (512 / 256 when memory-bound)
Calibration	16×256 tokens per language
Kept parts (Shrink)	8/16 head groups, 3,072/6,144 neurons
Seeds	0 (all runs); 1, 2 (main 4-layer runs)

Table 5: Training settings, identical for every trained method.

- **LaCo**: $\theta = \theta_A + \sum_{j \neq A}(\theta_j - \theta_A)$ for every weight; no data.

How the approaches differ: the control and AAM run the same function and differ in one switch (which layer of the pair is kept), so AAM − control measures the layer choice; PRISM-Shrink − control measures merging versus deleting. All share data, pairs, loss, optimizer and steps.

3.4.3 Parameters and architecture

One decoder layer with hidden size d, n_q query heads, n_{kv} key-value heads of size d_h and MLP width d_{ff} has

$$P_{\text{layer}} = 2\,d\,n_q d_h + 2\,d\,n_{kv} d_h + 3\,d\,d_{ff} + 2d + 2d_h, \qquad (10)$$

and the model has $P = Vd + L\,P_{\text{layer}} + d$ (tied embeddings, vocabulary $V = 151{,}936$).

Architectural differences: every final model is a standard Qwen3 model with fewer layers (same layer shape). During training only, PRISM-Shrink holds a double-width layer per pair (16 head groups, 6,144 neurons, plus a per-head q/k gain table of 5,632 parameters); AAM and the control hold the two original layers with a fade.

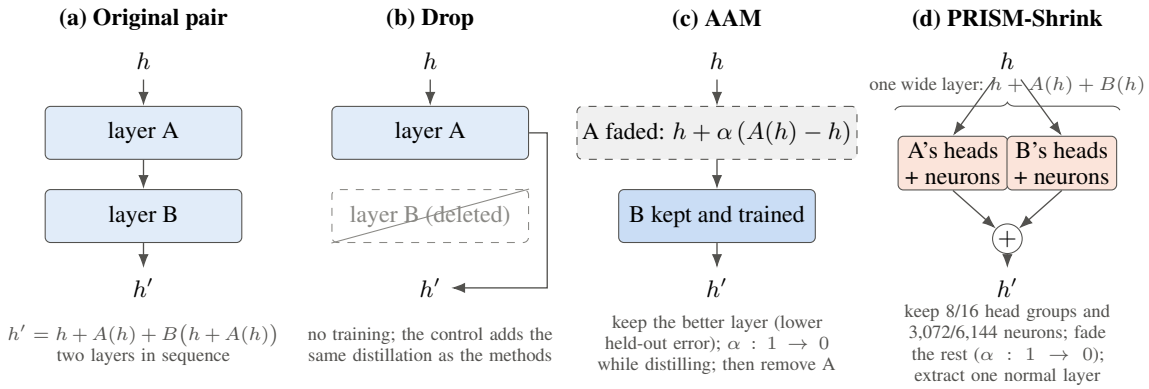


Figure 3: The four ways one pair of consecutive layers (A, B) is treated. Every final model has one standard layer per pair; AAM and PRISM-Shrink differ in what the remaining layer learns from.

Kept layer	Text	Steps	Acc	PPL	Copy
first (control)	plain	400	23.3	+25.2	0.579
first (control)	ICL	400	23.5	+24.5	0.372
better (AAM)	plain	400	**25.8**	+22.8	0.337
better (AAM)	ICL	400	25.1	+21.6	**0.246**
better (AAM)	plain	1000	**26.6**	+20.6	0.513
union (Shrink)	plain	400	24.3	+17.5	0.218
union (Shrink)	ICL	400	23.5	**+16.5**	**0.212**
union (Shrink)	plain	1000	24.0	**+15.8**	0.339

Table 6: Ablations, Qwen3-0.6B, 4 layers removed. Acc = few-shot average accuracy (%), PPL = mC4 perplexity increase (%), Copy = copying loss (lower is better).

3.4.4 Hyperparameters and ablations

No grid search: one Kaggle session per setting costs 1–3.5 GPU hours, so the recipe (learning rate, loss weight, fade length 0.7) was fixed from the small test and kept identical for all methods, which makes the comparison fair but means no method was tuned. Ablations: the kept layer moves accuracy by +1.6 to +2.5 points; the text by at most 0.8; 1,000 steps lower perplexity by about 2 points and change accuracy by at most +0.8 ($p = 0.05$, AAM) while copying gets worse (Table 6, Figure 7).

3.4.5 Metrics

- Perplexity $= \exp\big(-\frac{1}{T}\sum_t \log p(x_t \mid x_{<t})\big)$ on mC4 and Wiki-40B, reported as the increase over the original averaged over languages: it measures general language modelling damage.
- Few-shot accuracy on MMMLU, INCLUDE, MGSM, Belebele and XNLI (average of the five): it measures task ability. Random guessing gives 25 / 25 / 0 / 25 / 33.3 %, i.e. 21.7 % on the average.

Resource	Used
GPU	Kaggle, 2 × NVIDIA T4 (15 GB each)
Disk	Kaggle working directory (results < 1 GB)
Orchestration	local laptop (Kaggle API)
GPU time	21.0 session-hours logged

Table 7: Computational resources.

- Additional: skill kept above chance $= \frac{1}{5}\sum_b \frac{a_b - c_b}{a_b^{\text{orig}} - c_b}$; a copying probe (32 random 128-token sequences shown twice, NLL on the repeat; it targets induction heads, Olsson et al., 2022); unseen-language accuracy; prefill and decode speed; 95 % bootstrap confidence intervals and paired p-values.

3.4.6 System effort and demo scenario

Logged GPU time of the large-test queue: 21.0 session-hours on Kaggle (13 paired sessions including three that failed and were re-run; Table 14 in the appendix); earlier runs are not in this log. Each session uses two T4 GPUs (one notebook per GPU).

3.5 Project management

How the novelties were found: a layer analysis (angular change per layer, write sizes, update similarity, attention-head roles; Figure 14) showed that middle layers change the state little and read nearly the same input, which led to the parallel form (PRISM); the failure of closed-form fits on few-shot accuracy led to gradual distillation (AAM, PRISM-Shrink); the copying probe and head analysis led to ICL packing.

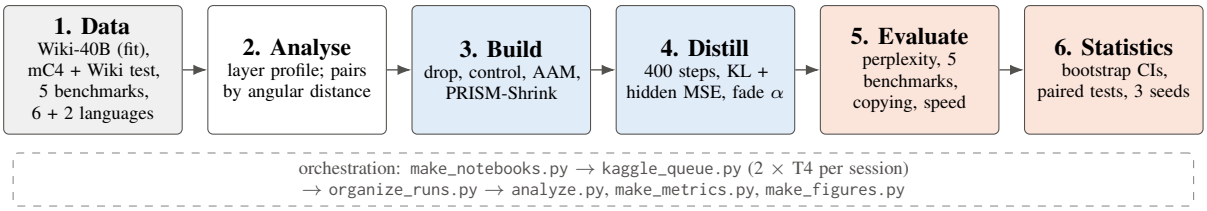


Figure 4: The end-to-end pipeline, from data to statistics, and the scripts that run it on Kaggle.

4 Experiment

4.1 Implementation

Code: https://github.com/sumitjhaa/gpt2-layermerging (branch `assignment3-top3`, folder `Assignment3/Sumit`). Pipeline:

1. Data preparation: download and cache Wiki-40B, mC4 and the five benchmarks; build calibration, training and test splits (shared by all notebooks through a reference run).
2. Layer and pair analysis; choose pairs by angular distance.
3. Build the method (`keep_one_build` for AAM and the control, `shrink_build` for PRISM-Shrink) and distill.
4. Evaluate: perplexity, five benchmarks, copying probe, speed; write one JSON row per model with every question's result.
5. Generate notebooks per experiment (`tools/make_notebooks.py`), run them on Kaggle (`tools/kaggle_queue.py`), file the runs (`tools/organize_runs.py`), and rebuild all tables and figures (`tools/analyze.py`, `make_metrics.py`, `make_figures.py`).

4.2 Results and findings

Step-by-step deductions:

1. *Deleting layers is very damaging and gets worse fast:* drop goes from +14 % perplexity (2 layers) to +689 % (8 layers); accuracy falls to chance from 4 layers (Table 8, Figure 5).
2. *Training alone repairs most perplexity damage but little accuracy:* the control cuts +84.7 % to +24.5 % at 4 layers but adds only +1.0 accuracy ($p = 0.05$). On 1.7B it adds +6.8.
3. *Choosing the better layer is what improves accuracy:* AAM beats the control by +4.8, +2.3, +1.5, +1.8 points at 2/4/6/8 layers (all $p \leq 0.018$) and by +3.2 on 1.7B (Figure 9); with the same text, the better layer gives +1.6 to +2.5 (Table 6). Three seeds give 25.8±0.3 (Table 10, Figure 12).
4. *Merging (PRISM-Shrink) protects perplexity and copying, not accuracy:* from 4 layers on, its perplexity is lower than the control's and AAM's in all 6 languages on both mC4 and Wikipedia, at every depth and on both models (Figure 11); its accuracy never differs significantly from the control (-1.0 to $+1.0$, $p \geq 0.07$). Figure 10 shows the resulting trade-off.
5. *Bigger models have more spare depth:* at 4 layers AAM keeps 41 % of the above-chance skill on 1.7B versus 21 % on 0.6B (Table 9, Figure 6).
6. *Maths needs the full depth:* MGSM falls from 41 % to about 2 % for every method at 4 layers (Table 11); drop and the control fall below chance on Belebele (about 20 %).
7. *Against quantization:* 8-bit quantization halves memory nearly without loss on Qwen3 and 4-bit loses about 5 MMLU points on Qwen3-8B (Zheng et al., 2025); removing 4 layers here saves 11 % and loses 23 points. Layer removal is only worth it for speed (here roughly 10–20 % faster at 4 layers) and can be combined with quantization.

4.3 Deployment

- Every compressed model is a standard Qwen3 model with fewer layers, so it can be saved with `save_pretrained` and loaded with Hugging Face `transformers` without custom code. The trained models were not saved in these runs; the notebooks rebuild them.
- The executable research pipeline already exists: one Kaggle notebook per experiment rebuilds the model, evaluates it and writes the results (Section 3.1); a 0.6B run takes about 1–1.5 hours on one T4.
- Inference needs about 1.2 GB (0.6B, fp16) or 3.4 GB (1.7B, fp16) of GPU memory for the weights; CPU inference is possible but slow.

Limitations

Removed	Method	Acc (%)	vs drop [95% CI]	Skill kept	mC4 PPL	Copy
0	original	48.9	–	100 %	–	0.125
2	drop	28.7	–	32 %	+14.2 %	0.320
2	control	29.3	+0.6 [−0.3, +1.5]	35 %	+7.9 %	0.172
2	AAM	**34.1**	**+5.4** [+3.9, +6.8]	**53 %**	**+6.0 %**	0.203
2	PRISM-Shrink + ICL	28.2	−0.4 [−1.5, +0.6]	31 %	+6.2 %	**0.131**
4	drop	22.6	–	7 %	+84.7 %	2.220
4	control	23.5	+1.0 [−0.0, +1.9]	13 %	+24.5 %	0.372
4	AAM	**25.8**	**+3.3** [+1.9, +4.6]	**21 %**	+22.8 %	0.337
4	PRISM-Shrink + ICL	23.5	+0.9 [−0.3, +2.3]	11 %	**+16.5 %**	**0.212**
6	drop	21.5	–	1 %	+236.6 %	4.292
6	control	22.3	+0.8 [−0.3, +2.0]	4 %	+59.2 %	0.506
6	AAM	**23.8**	**+2.3** [+1.0, +3.7]	**12 %**	+54.7 %	0.519
6	PRISM-Shrink + ICL	23.1	+1.6 [+0.4, +2.8]	7 %	**+39.4 %**	**0.340**
8	drop	20.9	–	−2 %	+689.4 %	10.112
8	control	21.6	+0.7 [−0.4, +1.9]	1 %	+103.8 %	1.091
8	AAM	**23.4**	**+2.5** [+1.1, +4.0]	**10 %**	+103.1 %	0.766
8	PRISM-Shrink + ICL	22.3	+1.4 [+0.3, +2.5]	4 %	**+69.0 %**	**0.651**

Table 8: Qwen3-0.6B-Base. Acc = few-shot average accuracy (random guessing 21.7 %); vs drop = paired difference on the same questions; skill kept = share of the original's above-chance accuracy; PPL = mean mC4 perplexity increase; Copy = copying loss.

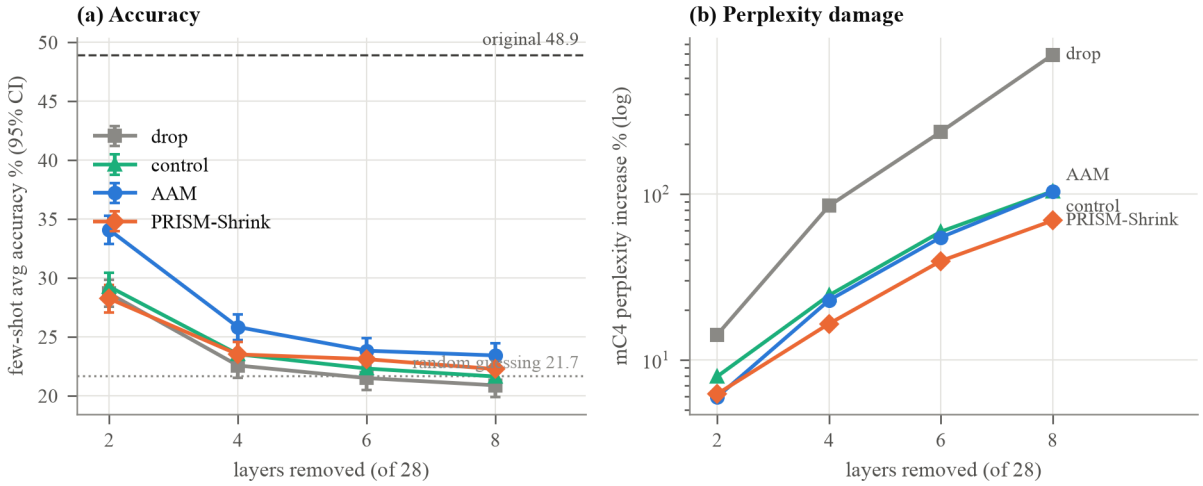


Figure 5: Accuracy (95% CI) and perplexity damage versus layers removed, Qwen3-0.6B.

Method	Acc	vs drop	PPL	Copy
original	58.9	–	–	0.106
drop	24.1	–	+70.5	2.831
control	30.9	+6.8	+32.1	1.150
AAM	**34.1**	**+10.0**	+29.4	0.450
PRISM-Shrink	31.5	+7.4	**+22.3**	**0.257**

Table 9: Qwen3-1.7B-Base, 4 layers removed (all methods trained on plain text). All differences vs drop $p < 0.001$.

Method	Acc (seeds 0/1/2)	Mean	vs control
AAM	25.8 / 25.4 / 26.1	25.8±0.3	+1.8 to +2.6*
PRISM-Shrink	23.5 / 23.7 / 23.5	23.6±0.1	−0.0 to +0.1
control	23.5 / 23.6 / 23.5	23.6±0.1	–

Table 10: Three seeds, Qwen3-0.6B, 4 layers removed. *significant on every seed ($p \leq 0.009$). Mean mC4 perplexity increase: AAM +23.0 %, PRISM-Shrink +16.6 %, control +24.2 %.

- At 4+ layers removed from 0.6B all methods are close to random guessing; maths is lost.
- PRISM-Shrink gives no accuracy gain, and its core idea is published (Xu et al., 2026).
- Only the main 0.6B 4-layer runs have three seeds; other settings are single runs.
- In the main 0.6B comparison AAM was trained on plain text and the control and PRISM-Shrink on ICL-packed text (the text ablation shows the effect is at most 0.8 points).
- One model family, two small sizes, six

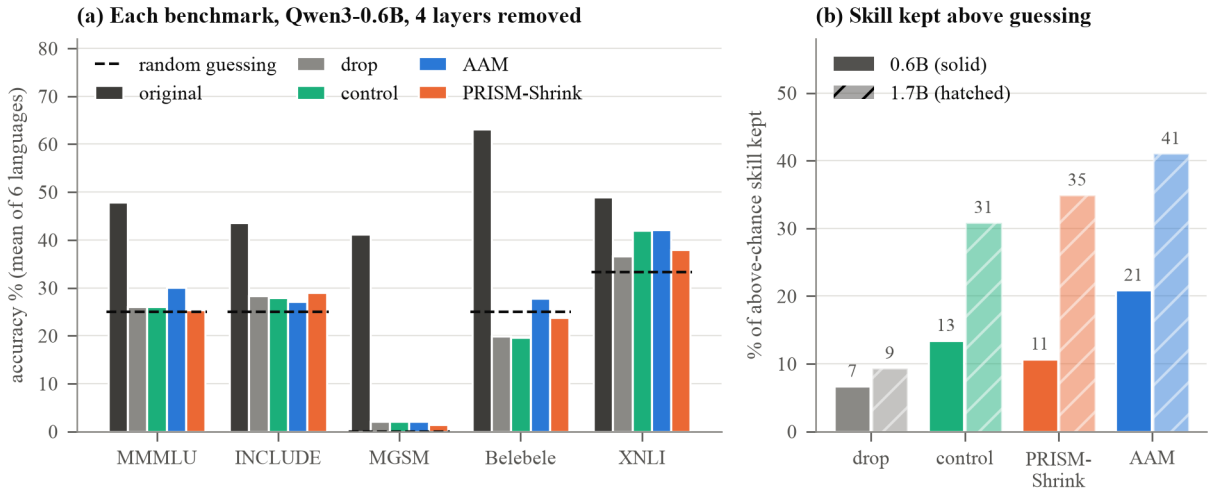


Figure 6: Left: each benchmark at 4 layers removed against random guessing (dashed). Right: share of the original's above-chance skill kept, 0.6B and 1.7B.

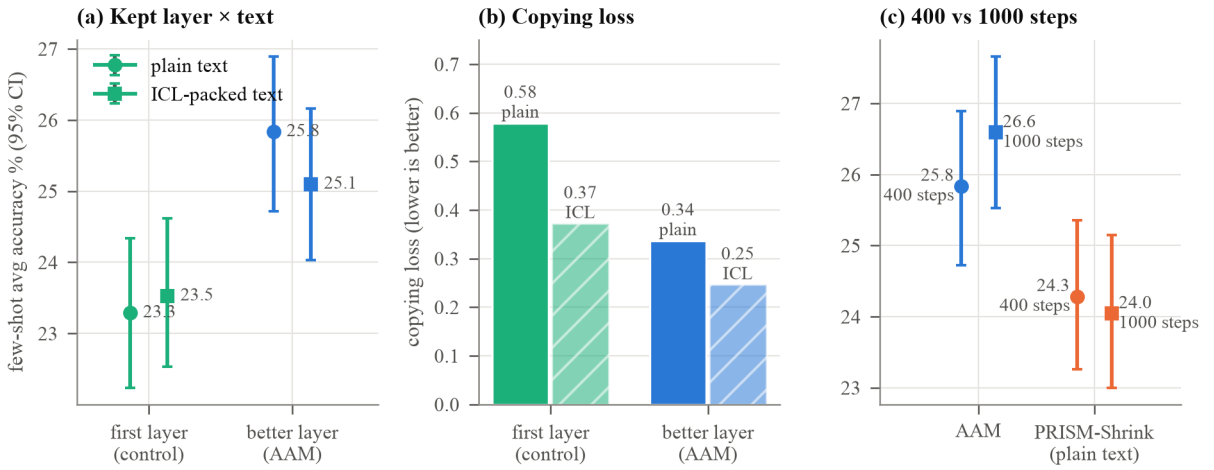


Figure 7: Ablations at 4 layers removed: kept layer × training text, copying loss, and training length.

languages, 200–250 questions per cell; no instruction-following or generation-quality tests beyond MGSM; hyperparameters not tuned per method.

- The quantization comparison is from the literature, not measured on this test; speed numbers are noisy (shared GPUs).
- Some earlier experiments keep only their numbers (code and raw results not committed); see `docs/limitations_and_issues.md`.

Team Contribution

Member	Contribution	Score
Sumit Kumar Jha	data pipeline, methods, implementation, experiments, analysis, report	100 %

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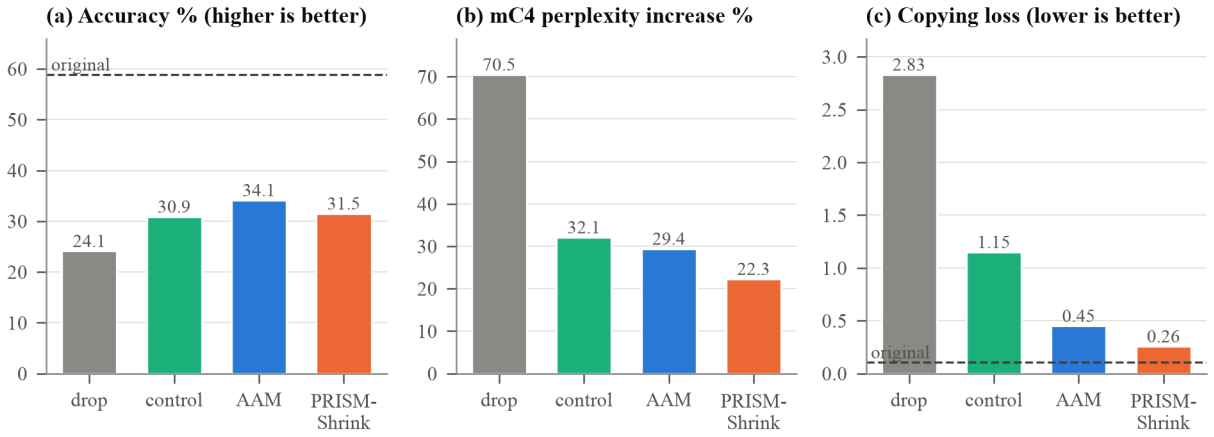


Figure 8: Qwen3-1.7B-Base, 4 layers removed.

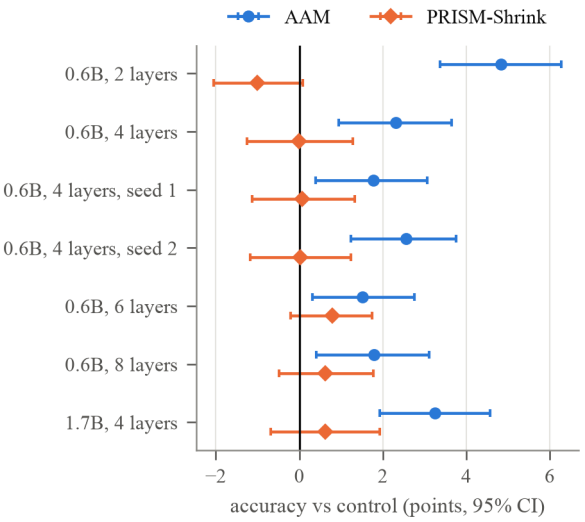


Figure 9: Is the gain real? Accuracy difference to the matched control on the same questions (paired bootstrap, 95% CI). AAM is right of zero in every setting; PRISM-Shrink overlaps zero in every setting.

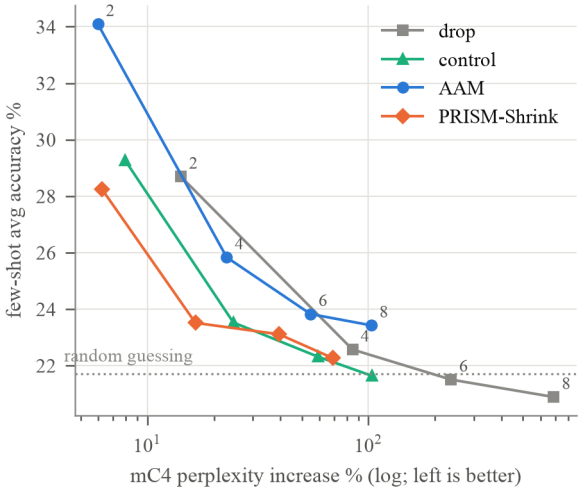


Figure 10: Accuracy against perplexity damage, Qwen3-0.6B; numbers are layers removed. AAM is highest, PRISM-Shrink is furthest left.

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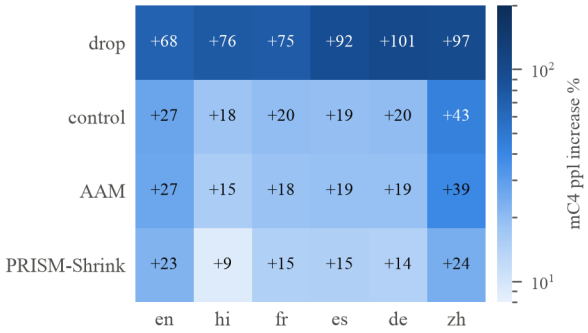


Figure 11: mC4 perplexity increase (%) per language, Qwen3-0.6B, 4 layers removed. Chinese is hurt most; PRISM-Shrink is lowest in every language.

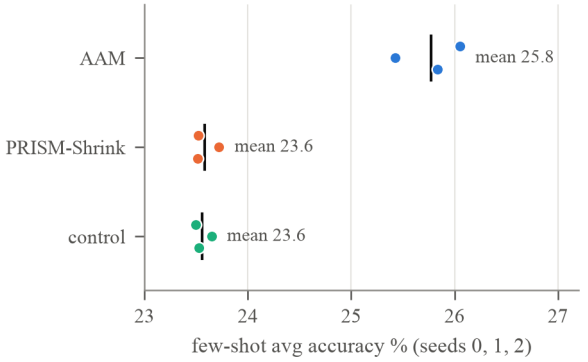


Figure 12: Three training seeds, Qwen3-0.6B, 4 layers removed; bars mark the means.

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A Additional results

Method	MMMLU	INCL.	MGSM	Bele.	XNLI
chance	25	25	0	25	33.3
original	47.9	43.6	41.2	63.1	48.9
drop	26.0	28.3	2.2	19.8	36.6
control	26.1	27.9	2.1	19.7	41.9
AAM	30.1	27.1	2.1	27.8	42.1
PRISM-Shrink	25.4	29.0	1.4	23.8	38.0

Table 11: Accuracy (%) per benchmark, Qwen3-0.6B, 4 layers removed, mean of the six languages.

Benchmark	ours 0.6B	official	ours 1.7B	official
MMLU (en)	54	52.8	59	62.6
MMMLU	47.9	50.2	50.9	63.3
INCLUDE	43.6	34.3	54.6	45.6
MGSM	41.2	31.0	58.8	50.7

Table 12: Original models on this test vs the Qwen3 report (Qwen Team, 2025), Table 8. Differences come from the languages (6 here), question subsets, prompts and MGSM shots (4 vs 8).

Language	Unique test tokens	Perplexity tokens
en	139,093	65,536
hi	330,752	65,536
fr	186,368	65,536
es	181,921	65,536
de	182,852	65,536
zh	156,288	65,536
ja (unseen)	197,929	65,536
ru (unseen)	168,363	65,536

Table 13: Size of the large test per language (Qwen3-0.6B tokenizer).

Session	Outcome	Hours
reference (0.6B)	complete	3.03
data ablation	complete	1.31
deep methods	error (Shrink OOM)	1.94
reference (1.7B)	error (dtype)	1.73
deep control + Shrink plain	complete	3.51
reference (1.7B), re-run	complete	0.77
methods (1.7B)	error (Shrink OOM)	1.12
Shrink deep 6 / 8	complete	1.47
Shrink + control (1.7B)	complete	1.09
1000 steps	complete	1.27
seeds AAM	complete	1.05
seeds Shrink	complete	1.40
seeds control	complete	1.34
Total		21.0

Table 14: Logged Kaggle GPU sessions of the large-test queue (two T4 GPUs each). Failed sessions were fixed and re-run; the AAM halves of the two out-of-memory sessions are used.

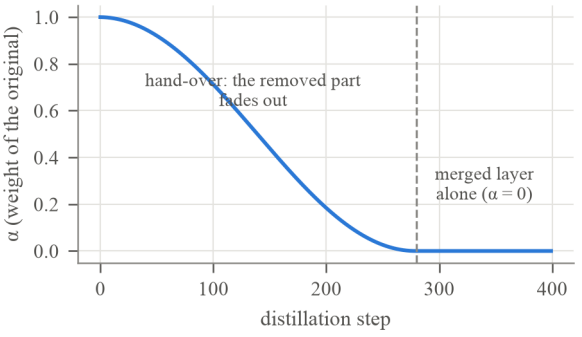


Figure 13: The fade schedule α over training steps.

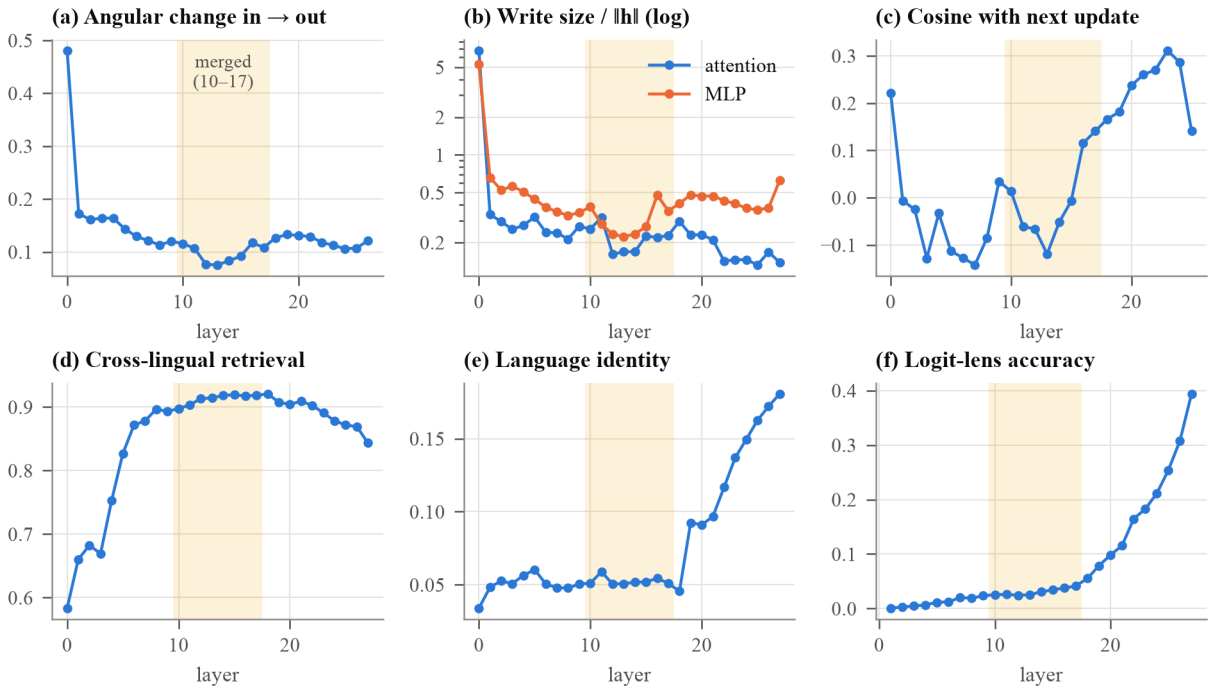


Figure 14: Layer analysis of Qwen3-0.6B (shaded: the layers merged at 4 layers removed). The middle layers change the state least (a) and write least (b); language identity and next-token readiness grow only in the last third (e, f).